Data Insight Challenge

Adaptive Assistive Cane (AAC)

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Introduction

Visual impairment is a significant issue for both global and national health. According to the World Health Organisation, over 2.2 billion people around the world are living with near or distance vision impairment. At least 1 billion cases of these impairments could have been prevented or have not yet been addressed. In the United States alone, there are approximately 7 million people with vision impairments, of which 1 million people have been classified as blind.

Safe and independent mobility is critical to maintaining quality of life as it allows people to navigate their daily environments, continue working, and participate in social activities. However, current assistive devices, such as traditional white canes that rely on physical contact, have limitations when it comes to proactive detection of obstacles and increasing the risk of falling or colliding with obstacles. Advanced smart canes often utilize sensors, GPS, or artificial intelligence (AI), but they are expensive and complex, making them inaccessible to many users.

To address this gap, this project aims to develop an assistive cane that uses ultrasonic sensors and a beeping system to provide real-time distance-based feedback to visually impaired users to support their travel safely. Ultimately, the goal of this project is to provide visually impaired individuals with increasing mobility safety through affordable, easy-to-use technology that bridges affordability and functionality.

Problem Statement

This uncertainty can lead to hesitance when walking and a reduction of confidence, which increases the risk of falling while trying to navigate.

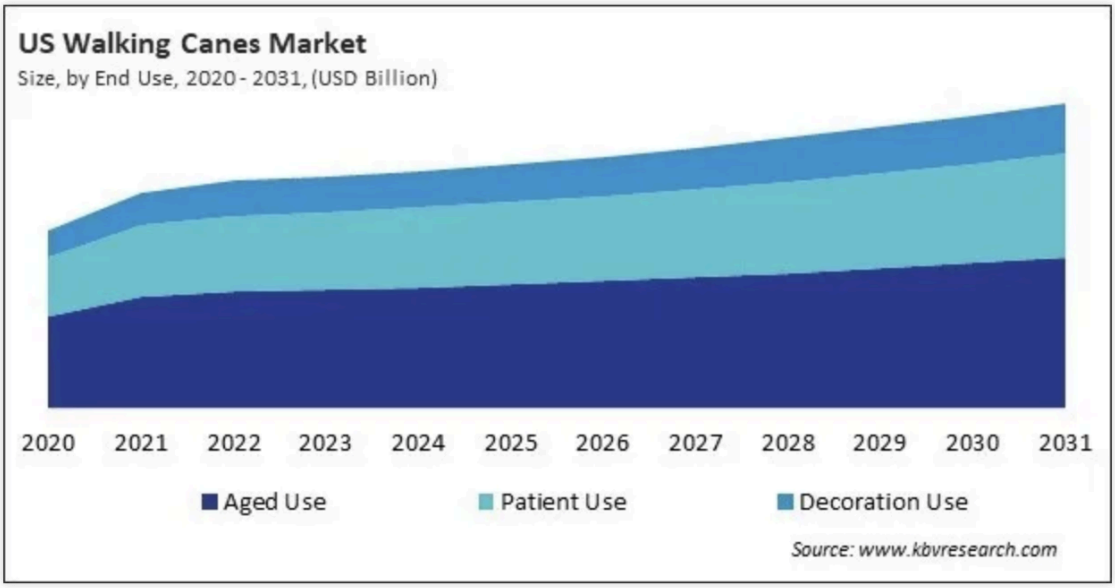
Traditional canes provide only basic tactile feedback when someone makes contact with an object (e.g., contact when using the cane to strike), and therefore they do not provide a warning about objects that are overhead, curbs, stairways or moving objects that may be in the person's path. More sophisticated assistive equipment (e.g., smart canes with ultrasonic sensors, navigation devices, or artificial intelligence) tend to be priced from $300–$1,000+, which limits

access for many users, especially poor individuals living in developing countries or persons on fixed income who are elderly.

Key Point: There is a clear gap between affordability and functionality. Most users are forced to choose between low-cost but limited tools or expensive high-tech solutions.

Data-Driven Justification

There are approximately 2.2 billion people worldwide with visual impairment, and further population growth and an aging population are anticipated to contribute to this problem, potentially resulting in more than 1.7 billion individuals having visual limitations by 2050. In the United States, rates of visual impairment are significantly higher for those adults over the age of 65, with more than 27% of people over the age of 71 having some form of visual impairment.



Diabetes and Vision Loss (Supporting Evidence)

As much as 26.4% of those who have diabetes have diabetic retinopathy (DR). An estimated 9.6 million people in the United States have DR, and about 1.84 million have eyesight affected by vision-threatening DR. DR contributes to many millions of cases of blindness and moderate-to-severe visual impairment globally and is projected to grow in relation to the increase in diabetes prevalence around the world.

Diabetic retinopathy can also lead to macular edema, hemorrhage, and eventual loss of vision. Individuals who experience vision loss will have increased dependence upon mobility devices. Chronic illnesses, such as diabetes, further complicate mobility challenges, as visual loss compounds other medical complications (i.e., neuropathy resulting in loss of balance).

Key Insights

- Increasing population at risk
- Greater demand for assistive mobility solutions
- Need for scalable and affordable technology
- The population at risk is increasing due to aging demographics and rising chronic illnesses.
- Greater demand exists for assistive mobility solutions that are both effective and scalable.
- There is a pressing need for affordable technology to prevent isolation, falls, and loss of independence.

Market Research & Gap Analysis

Existing Solutions

Traditional Canes

- Cost: $10–$40
- Features: Basic obstacle detection
- Limitation: No real-time feedback

Smart Canes

- Cost: $300–$1000+
- Features: Sensors, navigation, AI
- Limitation: High cost, complexity

Market Gap

Presently there exist only two choices in regards to finding a cane to meet your needs: inexpensive but simplistic (traditional) or expensive & high-tech (smart), lacking truly innovative mid-tier price points (e.g., < $100) providing critical real world data through pure functional components without added complexity. This represents a substantial unmet demand for developing new types of inexpensive devices that do include core safety features such as location-based audio notifications.

Proposed Solution

Product Overview - Adaptive Assistive Cane (AAC)

- Adaptive Assistive Navigation System
- Affordable smart cane with real-time feedback

We are proposing an Adaptive Assistive Cane (AAC), which is a low-cost (i.e., less than $50) smart sensory device that will attach to a walking stick in order to assist an individual with navigating their surroundings. By providing real-time data about the area surrounding the user through a combination of distance sensors and motion detection, the AAC is designed to be used with a standard walking stick.

The AAC differs from traditional smart canes that use one type of exposure sensor (i.e., ultrasound) by integrating multiple sensors into one device to increase the user's safety. This device will provide immediate feedback to the user when either navigating around obstacles or when detecting instability.

Affordability has been one of the design priorities of the AAC. It is constructed from commonly available electronic components and thus will be less than $50 to produce a prototype. When compared to currently available commercial alternatives, a device such as the AAC will be accessible to a larger population.

Core Features

- Distance-based obstacle detection

  Using distance data collected from ultrasonic sensors constantly measures the distance from the user to objects nearby so that obstacles can be detected before they reach the user.

- Range-Based Alert System
  - The user receives an alert using a buzzer, which will vary in intensity:

    - No sound or slow beep, indicating that the user is at a safe distance.
    - Beep frequency will start to increase as objects move closer to the user.
    - Constant alert when the user is in danger.
    - This will help the user intuitively recognize danger without needing visual cues.

- Adaptive sensitivity based on user motion

    The MPU6050 accelerometer detects movement and helps the system identify normal movements versus sudden movements that might indicate the user is unstable or may fall. This also helps the individual navigate where the stick is located.

- Behavior modes (idle, walking, danger proximity)

The system has four different operating modes including:

1. Idle - Normal activity with no users in range of the sensor.
2. Walking - Continuously monitors user's movement and beeps at moderate rate for close by obstacles.
3. Danger Proximity - High frequency sound when a user is near an obstacle.
4. Emergency - Very rapid sounds when a user is about to collide with an object.

Having these modes of operation enables easier use for customers by aligning how the device operates with their day-to-day environment.

Design Philosophy

- Simplicity - The system has been built with simplicity as a goal; it provides users with clear and easy-to-understand feedback without the need for complicated interaction or training.
- Accessibility - The system is designed to be as affordable as possible and is built from simple parts, so as to be made available to as many people as possible, including people living in resource-limited settings.
- Focus on essential safety features - The design prioritizes the basic needs of users by not adding unnecessary complication; the focus is on obstacle detection, awareness of proximity to objects and use of motion-based alerts.
- The design of the system ensures its dependable operation in the real world.

Data Analysis & Correlation

Methodology

- Data cleaning and preprocessing
- Selection of relevant variables (percent-based metrics)

The analysis of how diabetes relates to visual impairment was conducted with two data sets. One data set has the frequency of diabetes occurrence (diabetes prevalence), and the second data set gives the extent of sight threatening diabetic retinopathy (DR).

The initial step involved the collection and cleansing of the data. Given the two data sets utilized different formats to define age groups, it was necessary to standardize both data sets into three age groups with a common representation. The defined age categories are 18-44 years of age, 45-64 years of age and 65-74 years of age. This created a uniform definition of age across both data sets for accurate comparison purposes.

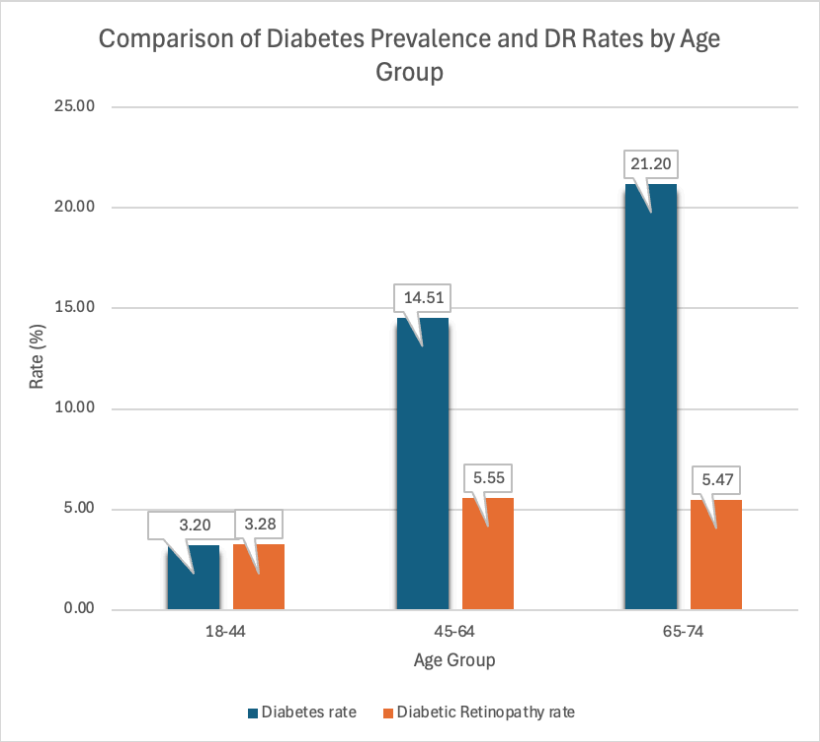
Secondly, all the external variables such as geographic location and demographic stratification were removed from both data sets for the purpose of concentrating only on the specific variables of diabetes prevalence and DR incidence in both data sets expressed as a percentage.

We then used pivot tables to determine the average values of diabetes prevalence and DR incidence for each of the three age groups of study. This representation of large data sets into a simplified and easily identifiable form allowed for subsequent analysis of the two data sets.

Finally, visualizations were created to better understand the patterns:

- A bar chart was used to compare diabetes and DR rates across age groups
- A scatter plot with a line of best fit was used to analyze the relationship between the two variables

This approach ensured that the analysis was both accurate and easy to interpret.



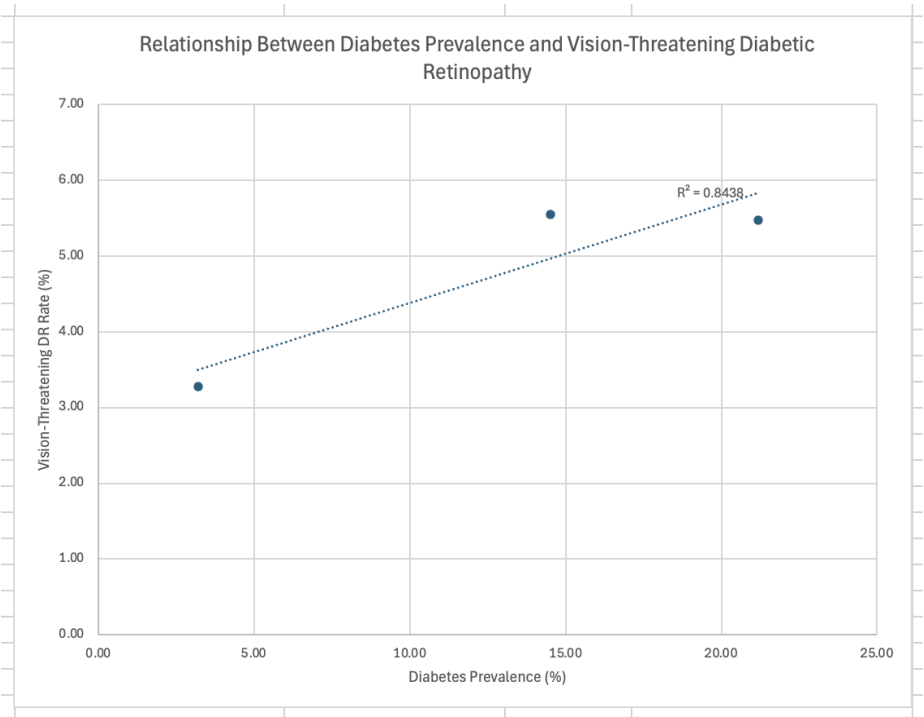
Correlation Analysis

The scatterplot demonstrates a strong correlation between the incidence of diabetes and the progression of diabetic retinopathy (DR), which can impact vision. The points shown on the scatterplot represent each age group and the rate of DR relative to the increasing incidence of diabetes.

A line of best fit has been plotted on the scatterplot to illustrate the overall relationship between the incidence of diabetes and the incidence of DR. The fact that the line moves in an upward direction indicates that the rate of incidence for DR increases as the incidence of diabetes increases.

The R-squared statistic of 0.84 for this relationship indicates a strong correlation (positive) between the two variables, as 84% of the variance in the rate of DR can be accounted for by variation in the incidence of diabetes. In summary, if someone has diabetes, they are likely to have complications that are related to vision later on in life.

While diabetes is a major factor for developing diabetic retinopathy (DR) as we age, other variables impacting the rate at which DR develops include time above target blood glucose levels, access to medical treatment, and lifestyle choices.



Findings

The analysis reveals several key insights:

- Diabetes rates in younger populations are roughly 3% while in older groups they are upwards of 21%.
- Although DR occurs more frequently as we age, the rate of increase for DR in older age groups is much slower than that of increase in diabetes.
- Relating diabetes to DR is supported by a strong positive correlation between increasing rates of diabetes and increasing risk of developing severe visual complications from DR.
- A slight plateau in DR rates, especially for the elderly suggests that the rate of progression for developing DR probably levels off sometime between 60-70 years old, or at least that it does so due to some type of medical intervention.

Overall, the data shows that diabetes is a major contributing factor to vision impairment, but not the only one.

Cost & Feasibility Analysis

- Microcontroller (Raspberry Pi Pico W) → ~$6–10
- Ultrasonic Sensor → ~$2–4
- MPU6050 Motion Sensor → ~$3–5
- Buzzer → ~$1–2
- LEDs (3 + resistors) → ~$1–2
- Breadboard + wires → ~$5–10
- Battery / power solution → ~$5–10
- Basic/standard cane → ~$10–20

Total Estimated Cost: ~$25–35 per unit

Target Market

- Visually impaired individuals
- Elderly population
- Individuals with chronic health conditions
- Caregivers and support systems

The market size is globally and nationally large because there are millions of people that are visually impaired, so there is a need for large numbers of people to access assistance for themselves, either in the more developed nations or because of the new and continuing developing countries.

Future Development

- Integration with Gemini AI
- Real-time voice navigation
- Mobile app connectivity
- Emergency alert systems

By adding these enhancements, it will change the product from a simple safety tool to an all-encompassing access system. The enhancements will add on to the affordability to everyone.

Impact & Conclusion

- Improved safety and mobility
- This project aims to enhance safety and independent mobility for visually impaired individuals through processes that give timely and intuitive feedback to help lower the risk of accidents and improve confidence.
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- Increased accessibility to assistive technology
- The project also enhances access to assistive technology for underserved populations through an affordable means of access to assistive technology through the technology gaps left by conventional and premium assistive technology.
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- Potential to scale and evolve
- Because of its scalability and ability to accommodate enhancements through the use of AI, the system can evolve to meet larger needs as they change over time.

The Adaptive Assistive Cane is bridging the gap between affordability and functionality while providing solutions to a growing need created by the ever-increasing number of retirees and the increase in chronic conditions, such as diabetes. This innovation will not only enhance the quality of life for consumers of the product but also illustrates how accessible technology can make a positive impact on society.

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